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BRAKE PLATE FOR A ROPE GRAB FOR ROPE CLIMBING, ROPE GRAB FOR ROPE CLIMBING, AND METHOD FOR RETROFITTING A ROPE GRAB

Abstract

The invention relates to a brake plate for a rope grab for rope climbing, comprising: a first section for supporting a rope on the brake plate, said first section being a first arc of a first circle with a first radius, and a second section for supporting the rope on the brake plate, said second section being arranged adjacently to the first section, wherein the brake plate is characterized in that the second section is a second arc of a second circle with a second radius, and the second radius is larger than the first radius. The invention additionally relates to a rope grab for rope climbing, comprising the brake plate, and to a method for retrofitting a rope grab with the brake plate.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. patent application Ser. No. 17/779,603, entitled “BRAKE PLATE FOR A ROPE GRAB FOR ROPE CLIMBING, ROPE GRAP FOR ROPE CLIMBING, AND METHOD FOR RETROFITTING A ROPE GRAB,” filed May 25, 2022, which claims priority to International Patent Application No. PCT/EP2020/085412, filed Dec. 10, 2020, and European Patent Application Serial No. 19215073.8, filed Dec. 11, 2019, each of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The invention described herein relates to a brake plate for a rope grab for rope climbing. Furthermore, the invention relates to a rope grab for rope climbing comprising the brake plate, and to a method for retrofitting a rope grab with the brake plate.

BACKGROUND

[0003] A rope grab known from the prior art is shown in FIGS. 1A to 1C. This prior art rope grab has a circular brake plate 12 and a circular control plate 13 along which or around which the rope 50 passes through the rope grab. The brake plate 12 and the control plate 13 thus each have the contour or outline of a circle. The rope 50 has a first rope end 18, which is secured to the belayer who normally stands on the ground. The rope 50 further has a second rope end 11 which is secured to the climber. The second rope end 11 is thus the rope end which is pulled and can be tautened and is therefore referred to as the load strand. The first rope end 18, on the other hand, is the loose, non-pulled, slack rope end and is therefore referred to as the slack strand. The rope grab further typically includes a safety pin 14 and a cheek 15 having an opening 15a for passing a carabiner 17 therethrough, wherein the carabiner 17 in turn is secured to a lug 16 secured to a (rock) wall 60. The rope grab is usually suspended from an intermediate belay.

[0004] In principle, this rope grab can be in three positions, namely the rest position shown in FIG. 1A and the work positions 1 and 2 shown in FIGS. 1A and 1B. In the rest position, the rope 50 runs through the rope grab without load and there is little friction on the brake plate 12 and the control plate 13. While in the work position 1, the rope 50 still runs through another carabiner located above the rope grab on the wall 60, this is not the case in the work position 2, and in the work position 2, the rope grab serves as a deflection means as shown in FIG. 1C.

[0005] A climber's fall in rope climbing can basically be divided into three phases. In the first phase, the climber loses his grip and begins to fall. The belayer possibly notices the fall of the climber and prepares for the impact force. In the second phase, the rope grab 50 tightens between the climber and the belayer and between the brake plate 12 and the control plate 13. The rope grab turns in between the brake plate 12 or control plate 13 and the rope 50 via the normal force (see FIG. 1B). The rope grab is pulled along until the intermediate belay, such as carabiner 17 and a lug 16, retains the rope grab. The belayer is slightly carried along by the kinematic energy from the climber, and the rope elongation serves as a damping means for the climber and belayer. This is the moment of the impact force. The belayer is accelerated to the speed of the falling climber. In the third phase, kinetic energy is converted via friction in the rope grab system into heat and possibly other forms of energy such as potential energy of the belayer, and the falling climber is decelerated. [0006] Due to the rope grab turning from the rest position into the work position (see FIG. 1B), the

slack rope is shortened. Through the tension from the rope, the rope grab is pulled into the work position and a portion of the impact force can already be decoupled from the rope in the form of friction via the rope grab. A portion of the forces acting in the rope **50** are decoupled via the friction of the rope grab until the system has come to a standstill.

[0007] However, it has been shown that certain greater weight differences between a heavy climber and a lighter belayer result in the fact that during a fall, a climber can be arrested by such rope grabs only with great difficulties.

SUMMARY

[0008] It is an object of invention to find means which make it possible that during rope climbing, climbers having an even greater mass in comparison to the belayer can be secured.

[0009] This object is achieved in that the brake plate according to the invention is used instead of the circular brake plate **12**. The brake plate according to the invention can be used in a rope grab known in the prior art, such as the rope grab described in FIGS. **1A** to **1C**. Therefore, the object is achieved in each case by the brake plate according to the invention, the rope grab according to the invention and the method according to the invention.

[0010] The brake plate according to the invention for a rope grab for rope climbing comprises: a first section for supporting a rope on the brake plate, wherein the first section is a first arc of a first circle having a first radius; and a second section for supporting the rope on the brake plate, wherein the second portion is arranged adjacent the first portion; wherein the brake plate is characterized in that the second section is a second arc of a second circle having a second radius, wherein the second radius is greater than the first radius.

[0011] The rope grab for rope climbing according to the invention is characterized in that the rope grab comprises the brake plate according to the invention.

[0012] The method according to the invention is for retrofitting a rope grab, wherein the rope grab comprises a fastening means for a brake plate, the method comprising: fixing the brake plate according to the invention to the fastening means.

[0013] Accelerations acting on the belayer become smaller when the rope grab responds faster. By means of a lever arm, which is provided between the pivot point of the rope grab and the driven brake plate, a moment is built up that turns the rope grab into the rope course. This turn takes the slack rope out of the system and the wrap angles of the rope at the brake plate and the control disc increase (compare FIG. **1A** with FIG. **1B**, **1C**). The larger this lever arm can be kept, the more the rope grab is pulled into the rope course. According to the brake plate according to the invention, the lever arm does not decrease so much or so fast during the turning of the rope brake compared to the circular brake plate **12** known from the prior art, so that the rope grab can be turned in faster and farther and the wrap angle of the rope around the brake plate and control plate increases, so that as a result, greater weights or weight differences can be arrested by the rope brake when the climber falls. Due to the brake plate according to the invention, more force is decoupled from the rope and less energy from the impact of the fall reaches the belayer, who is also less accelerated as a result. From the belayer's point of view, the fall is thus more controllable.

[0014] According to preferred embodiments of the brake plate according to the invention, the brake plate further has a third section for supporting the rope on the brake plate, wherein the third section is a third arc of a third circle having a third radius, wherein the third section is arranged adjacent to the first section, and wherein the third radius is smaller than the first radius, thereby further enhancing the positive effects described above.

[0015] According to preferred embodiments of the brake plate according to the invention, the second section is a section of the surface of the brake plate and has a rope channel with wavelike contour, whereby the rope adapts to the brake plate during the climber's fall and the friction values are additionally increased, in particular in the work position **2** (FIG. **1C**).

[0016] These optional preferred features can be combined with each other. Further optional

preferred features are described in the following part of the description, resulting in further advantages for the person skilled in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1A shows the rest position of a rope grab known from the prior art;

[0018] FIG. 1B shows a first work position of the rope grab known from the prior art;

[0019] FIG. 1C shows a second work position of the rope grab known from the prior art;

[0020] FIG. 2 shows the brake plate **10** according to the invention in a state in which it is installed in the rope grab **100** according to the invention and, as an example, is in the work position according to FIG. 1C;

[0021] FIG. 3 shows the brake plate **10** according to the invention;

[0022] FIG. 4 shows the brake plate **10** according to the invention in a preferred embodiment;

[0023] FIG. 5 shows the brake plate **10** according to the invention in a preferred embodiment of the brake plate **10** as shown in FIG. 3;

[0024] FIG. 6A shows a front view of the brake plate **10** according to the invention in the preferred embodiment as shown in FIG. 5; and

[0025] FIG. 6B shows a vertical section through the embodiment of the brake plate according to the invention in the preferred embodiment according to FIG. 2A along the axis A-A shown in FIG. 2A.

[0026] Features with the same reference signs in FIGS. 1A to 6B are identical, so that the features described in the context of FIGS. 1A to 1C are equally applicable to FIGS. 2 to 6B, provided they are used there with the same reference signs.

DETAILED DESCRIPTION

[0027] FIG. 3 shows the brake plate **10** according to the invention for a rope grab **100** for rope climbing. FIG. 2 shows the brake plate **10** according to the invention in a state in which it is installed in the rope grab **100** according to the invention and, as an example, is in the work position according to FIG. 1C.

[0028] The brake plate comprises: a first section **1** for supporting a rope **50** on the brake plate **10**, wherein the first section **1** is a first arc of a first circle Ci1 having a first radius R1; and a second section **2** for supporting the rope **50** on the brake plate **10**, wherein the second section **2** is arranged adjacent to the first section **1**; wherein the brake plate **10** is characterized in that the second section **2** is a second arc of a second circle Ci2 having a second radius R2, wherein the second radius R2 is greater than the first radius R1, wherein preferably the ratio of the second radius to the first radius is (approximately) 1 to ½. “Adjacent” in the meaning of this application means having a common boundary or directly merging, abutting each other, or touching each other.

[0029] The first section **1** and the second section **2** can be sections of the surface of the brake plate **10**, and the second section **2** can be located downstream of the first section **1** as viewed from the slack strand **18** of the rope to the load strand **11** of the rope. The second rope end **11** is the rope end that is pulled and can be tightened and is therefore referred to as the load strand. The first rope end **18**, on the other hand, is the loose, non-pulled, slack strand end of the rope and is therefore referred to as the slack strand. The first rope end **18** can also be referred to as the rope **50** entering the rope grab **100** or brake plate **10**, and the second rope end **11** can also be referred to as the rope **50** exiting the rope grab **100** or brake plate **10**. The rope end **11** is for attachment to climber and the rope end **18** is for attachment to belayer. The rope grab usually has a further plate **13** (control plate) and the brake plate **10** and the further plate are arranged within the rope grab **100** in such a manner that the rope **50** can be fed into the rope grab **100** from the direction of the belayer (slack strand) via the further plate **13**, is guided onto the first section **1** of the brake plate and is deflected at the brake

plate **10** to exit the rope grab **100** in the direction of the climber (load strand).

[0030] The first arc and the second arc can be arranged adjacent to each other and are preferably connected by a tangent **25a**, **25c**.

[0031] As shown in FIG. **4** in a preferred embodiment, the brake plate **10** can further comprise a third section **3** for supporting the rope **50** on the brake plate **10**, wherein the third section **3** is a third arc of a third circle **Ci3** having a third radius **R3**, wherein the third section **3** is arranged adjacent the first section **1** and wherein the third radius **R3** is smaller than the first radius **R1** and the second radius **R2**, respectively, wherein for the effectiveness of the brake plate, the ratio of the second radius to the first radius to the third radius is particularly preferably (approximately) 1 to $\frac{1}{2}$ to $\frac{1}{3}$.

[0032] The third section **3** can be a section of the surface of the brake plate **10**, and the first section **1** can be located downstream of (and adjacent to) the third section **3** as viewed from the slack strand **18** of the rope to the load strand **11** of the rope **50**. The first arc and the third arc can be arranged adjacent to each other and are preferably connected by a tangent.

[0033] The first section **1**, the second section **2** and optionally the third section **3** can be referred to as the front side of the brake plate **10**, while the fourth section **4** shown in FIGS. **2** to **6B** can be referred to as the rear side of the brake plate **10**. The shape of section **4** can be arbitrary. The rope **50** runs into the rope grab **100**, optionally without contacting the fourth section **4**, and runs out of the rope grab **100** via the sections **1**, **2** and optionally **3** of the brake plate **10** on which the rope **50** rests.

[0034] In each of the embodiments described by FIGS. **2** to **6B**, the centers **C1**, **C2** and **C3**, respectively, of the circles **Ci1**, **Ci2** and **Ci3**, respectively, can be arranged spatially separated from each other. The arrangements of the center points shown in the figures are examples but not essential to the invention.

[0035] In any of the embodiments described by FIGS. **2** to **4**, the second section **2** can be a section of the surface of the brake plate **10** and can have a rope channel **20** with a wave-like contour, as also shown and described in FIGS. **5** to **6B**.

[0036] FIG. **5** shows the brake plate **10** according to the invention as shown in FIG. **3**, additional with rope channel **20**.

[0037] FIG. **6A** shows a front view of the brake plate **10** according to the invention in the preferred embodiment according to FIG. **5**. FIG. **6B** shows a vertical section through the embodiment of the brake plate according to the invention in the preferred embodiment according to FIG. **2A** along axis A-A shown in FIG. **2A**.

[0038] As shown in FIGS. **5** to **6B**, the rope channel **20** can have one or more (any number of) recesses **21a**, **21b**, **21c** on the surface of the brake plate **10**, wherein each of the recesses **21a**, **21b**, **21c** can be delimited by two constrictions **22a**, **22b**, **22c**, **22d**. The rope channel **20** can further include one or more recesses which are delimited by a constriction **22a**, **22d** on one side only and which end in the surface of section **2** of the brake plate **10** at another end.

[0039] The one or more recesses **21a**, **21b**, **21c** with the respective constrictions **22a**, **22b**, **22c**, **22d** arranged adjacent thereto can be adapted here in such a manner that, when loaded, the rope structure or the sheath of the rope **50** supported in the rope channel **20** is compressed and the section of the rope **50** with the compressed rope structure or the compressed rope sheath is pressed into one or more of the recesses **21a**, **21b**, **21c**.

[0040] In this case, the one or more recesses **21a**, **21b**, **21c** can comprise a first recess **21a**, **21b** and a second recess **21b**, **21c**, wherein, as viewed from the slack strand **18** of rope **50** to the load strand **11** of rope **50**, the second recess **21b**, **21c** is arranged downstream of and preferably adjacent to the first recess **21a**, **21b** (as in the case of recess **21b** with respect to recess **21a** or in the case of recess **21c** with respect to recess **21b**) on the surface of the brake plate **10**, and between the first recess **21a**, **21b** and the second recess **21b**, **21c** there is one (**22b** or **22c**) of the constrictions **22a**, **22b**, **22c**, **22d** which delimits the first recess **21a** from the second recess **21b**, **21c**.

[0041] On the side of each of the constrictions **22a**, **22b**, **22c**, **22d** facing the slack strand **18** of the rope **50**, the rope **50** can run onto the constriction, and on the side of each of the constrictions **22a**, **22b**, **22c**, **22d** facing the load strand **11** of the rope **50** it can run off the constriction.

[0042] The one or more recesses **21a**, **21b**, **21c** with the respective constrictions **22a**, **22b**, **22c**, **22d** arranged adjacent thereto can be adapted here in such a manner that, when loaded, the rope structure or the sheath of the rope **50** supported in the rope channel **20** is more compressed within a first one of the recesses **21a**, **21b**, **21c** and the compressed rope **50** is more pressed or pulled into the recess than within a second one of the recesses **21a**, **21b**, **21c**.

[0043] When the climber falls, the rope **50** is pulled into the wave-like rope channel **20**. Due to friction on the flanks of the one or more recesses in the rope channel **20**, the rope structure is deformed or compressed or the rope sheath is pushed back or compressed with respect to the rope core. The section of the rope **50** with a deformed rope structure or a pushed-back or compressed rope sheath is pressed into one or more of the recesses **21a-c** in the rope channel and thereby increases the frictional force acting on the rope **50**.

[0044] Both the contour **23a** of the first recess **21a** and the contour **23b**, **23c** of the second recess **21b**, **21c** can each have the shape of a surface section of an ellipsoid, wherein a first angle α_1 is given between a major axis **24a** of the ellipsoid of the first recess **21a** and a tangent **25a** to the second circular arc in the region of the first recess **21a**, wherein a second angle α_2 is given between a major axis **24c** of the ellipsoid of the second recess **21c** and a tangent **25c** to the second circular arc in the region of the second recess **21c**, and wherein the first angle α_1 is greater than the second angle α_2 , wherein for the effectiveness of the brake plate, the first angle and the second angle are each particularly preferably between 10° and 90° .

[0045] Since the major axes **24a**, **24c** run parallel to one another, the angle α between the tangent **25a**, **25c** and the major axis **24a**, **24c** changes in such a manner that the slope which the constrictions **22a-d** present to the incoming rope **50** is initially steep and then becomes increasingly shallower and as a result, the rope **50** is initially pressed strongly into the respective recess **21a-c** along the second section **2** in the direction from the slack strand to the load strand and then, as viewed in the direction of the load strand, is drawn less and less into the respective recess **21a-c**. When the rope **50** is under load, this results in an increased frictional force acting on the rope **50**. If the rope **50** is free of load, however, this allows the rope **50** to be freely lifted out of the rope channel **20** again.

[0046] In this regard, the rope grab **100** according to the invention for rope climbing is characterized in that the rope grab **100** comprises the brake plate **10** according to any one of the embodiments described herein.

[0047] The method according to the invention is for retrofitting a rope grab brake **100**, wherein the rope grab **100** comprises a fastening means **30** (e.g., a screw) for a brake plate, the method comprising: fixing the brake plate **10** according to the invention according to any one of the embodiments described herein to the fastening means **30**. The brake plate according to the invention can comprise an opening **40** for fixing to the fastening means **30**. The brake plate according to the invention can be manufactured from a metal.

[0048] The numerical values mentioned herein are examples but not essential to the invention.

[0049] The detailed description and the drawings describe preferred embodiments of the invention given by the following claims. For carrying out the invention given herein according to the appended claims, the (optional or preferred) features disclosed in the foregoing description, claims and drawings can be used both individually and in any combination in configurations of the invention.

Claims

- 1.** A brake plate for a rope grab for rope climbing, the brake plate comprising: a first section for supporting a rope on the brake plate, wherein the first section is a first arc of a first circle having a first radius; and a second section for supporting the rope on the brake plate, wherein the second section is arranged adjacent to the first section; wherein the brake plate is configured such that the second section is a second arc of a second circle having a second radius, wherein the second radius is greater than the first radius, the first section and the second section are sections of a surface of the brake plate, and the second section is arranged downstream of the first section as viewed from a slack strand of the rope to a load strand of the rope, the brake plate further comprises a third section for supporting the rope on the brake plate, wherein the third section is a third arc of a third circle having a third radius, wherein the third section is arranged adjacent to the first section and wherein the third radius is smaller than the first radius, and the third section is a section of the surface of the brake plate and the first section is arranged downstream of the third section as viewed from the slack strand of the rope to the load strand of the rope.
- 2.** The brake plate of claim 1, wherein the first arc and the second arc are arranged adjacent to each other and are connected to each other by a tangent.
- 3.** The brake plate of claim 1, wherein the centers of the circles are arranged spatially separated from each other.
- 4.** The brake plate of claim 1, wherein the second section is a section of the surface of the brake plate and has a rope channel with a wavelike contour.
- 5.** The brake plate of claim 4, wherein the rope channel has one or more recesses on the surface of the brake plate, wherein each of the recesses is delimited by two constrictions.
- 6.** The brake plate of claim 5, wherein the one or more recesses with the respective adjacent constrictions are adapted such that when loaded, a rope structure or a sheath of the rope supported in the rope channel is compressed and the section of the rope with the compressed rope structure or the compressed rope sheath is pressed into one or more of the recesses.
- 7.** The brake plate of claim 6, wherein the one or more recesses with the respective adjacent constrictions are adapted such that when loaded, the rope structure or the sheath of the rope supported in the rope channel is compressed more within a first one of the recesses and the compressed rope is pressed more into the recess than within a second one of the recesses.
- 8.** The brake plate of claim 7, wherein both the contour of the first recess and the contour of the second recess each have the shape of a surface portion of an ellipsoid, wherein a first angle is given between a major axis of the ellipsoid of the first recess and a tangent to the second arc in the region of the first recess, wherein a second angle is given between a major axis of the ellipsoid of the second recess and a tangent to the second arc in the region of the second recess, and wherein the first angle is greater than the second angle.
- 9.** The brake plate of claim 5, wherein the one or more recesses comprise a first recess and a second recess, wherein, as viewed from the slack strand of the rope to the load strand of the rope, the second recess is arranged downstream of the first recess on the surface of the brake plate, and between the first recess and the second recess there is one of the constrictions which delimits the first recess from the second recess.
- 10.** The brake plate of claim 5, wherein the rope runs onto the constriction on the side of each of the constrictions facing the slack strand of the rope and runs off the constriction on the side of each of the constrictions facing the load strand of the rope.
- 11.** A rope grab for rope climbing, the rope grab comprising: a brake plate, the brake plate comprising: a first section for supporting a rope on the brake plate, wherein the first section is a first arc of a first circle having a first radius; and a second section for supporting the rope on the brake plate, wherein the second section is arranged adjacent to the first section; wherein the brake plate is configured such that the second section is a second arc of a second circle having a second radius, wherein the second radius is greater than the first radius, the first section and the second

section are sections of a surface of the brake plate, and the second section is arranged downstream of the first section as viewed from a slack strand of the rope to a load strand of the rope, the brake plate further comprises a third section for supporting the rope on the brake plate, wherein the third section is a third arc of a third circle having a third radius, wherein the third section is arranged adjacent to the first section and wherein the third radius is smaller than the first radius, and the third section is a section of the surface of the brake plate and the first section is arranged downstream of the third section as viewed from the slack strand of the rope to the load strand of the rope.

12. A method for retrofitting a rope grab, wherein the rope grab comprises a fastening means for a brake plate, the method comprising: fixing the brake plate to the fastening means, the brake plate comprising: a first section for supporting a rope on the brake plate, wherein the first section is a first arc of a first circle having a first radius, and a second section for supporting the rope on the brake plate, wherein the second section is arranged adjacent to the first section, wherein the brake plate is configured such that the second section is a second arc of a second circle having a second radius, wherein the second radius is greater than the first radius, the first section and the second section are sections of a surface of the brake plate, and the second section is arranged downstream of the first section as viewed from a slack strand of the rope to a load strand of the rope, the brake plate further comprises a third section for supporting the rope on the brake plate, wherein the third section is a third arc of a third circle having a third radius, wherein the third section is arranged adjacent to the first section and wherein the third radius is smaller than the first radius, and the third section is a section of the surface of the brake plate and the first section is arranged downstream of the third section as viewed from the slack strand of the rope to the load strand of the rope.
